WEB DEVELOPMENT (INTRODUCTION)

WEDE5020

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Part 1: Website Structure and Planning

Des-Lux Play

Updated to reflect the finished Part 3 website (October 2026)

Revision note

This document was written in Part 1 and has been brought up to date so that it matches the finished website.

Section	What was updated
1. Sitemap	New diagram and hierarchy that include the supporting files (404.html, robots.txt, sitemap.xml) and the features of each page.
2. File and Folder Structure	The real structure: six JavaScript files, docs/, WebP images, files at the repository root.
3. Page Structure Descriptions	Each page's structure now includes its Part 3 components (catalogue, tabs, modal, gallery, accordion, map, result panels). Added a Part 3 elements list.
4. Navigation System	The search box now works, the footer repeats the menu, current-page marking and skip link added.
5. References	Original references kept; Part 3 sources added.

Table of Contents

1. Sitemap

2. File and Folder Structure

3. Page Structure Descriptions

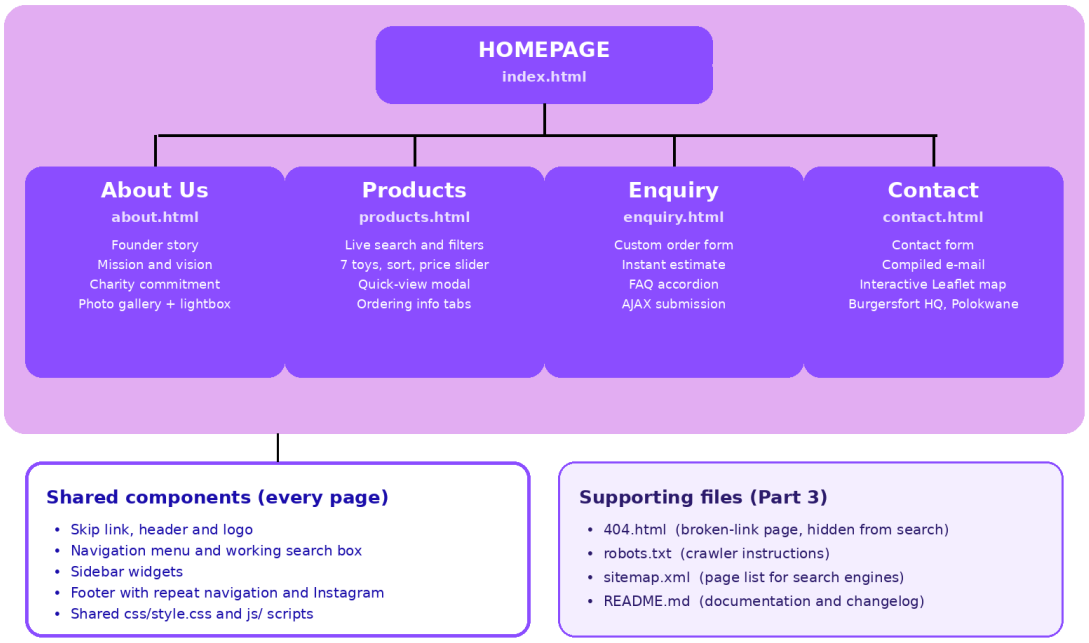
4. Navigation System

5. References

1. Sitemap

The following sitemap illustrates the hierarchical structure of the Des-Lux Play website. The site consists of five core pages, with the homepage serving as the central hub from which all other pages are accessible. Part 3 added supporting files for search engines and a page for broken links.

Visual Sitemap:



Original Canva sitemap (Part 1): https://canva.link/uy2ftq4i8oas7rt

Hierarchy Description

```
Level 0 (Root): Homepage (index.html)
    +-- Level 1: About Us (about.html)
    +-- Level 1: Products (products.html)
    +-- Level 1: Enquiry (enquiry.html)
    +-- Level 1: Contact (contact.html)

Supporting files (not in the menu):
    +-- 404.html      shown for broken links, hidden from search
    +-- robots.txt    tells search engines what to crawl
    +-- sitemap.xml   lists the five pages for search engines
```

The sitemap follows a flat hierarchy where all secondary pages are directly accessible from the homepage. This ensures that users can reach any section of the website within a single click, improving usability and reducing navigation complexity. Each page contains a consistent navigation menu, allowing users to move between pages without returning to the homepage.

Pages also link to each other in context: the homepage collection buttons open the filtered catalogue (for example products.html?category=limited), About links to the catalogue and the enquiry form, and each custom toy in the catalogue links to the enquiry form with that toy already selected. These links help visitors and strengthen internal linking for search engines. The sitemap.xml file is a separate machine-readable list for search engines and is not part of the visitor navigation.

2. File and Folder Structure

A well-organised file and folder structure is essential for maintaining a scalable and manageable website. The following structure is the one implemented for the finished Des-Lux Play project.

```
des-lux-play/
|
+-- index.html            (Homepage)
+-- about.html            (About Us page)
+-- products.html         (Products catalogue)
+-- enquiry.html          (Custom order enquiry form)
+-- contact.html          (Contact page with two locations)
+-- 404.html              (Page not found)
+-- robots.txt            (Search engine crawl rules)
+-- sitemap.xml           (Page list for search engines)
+-- README.md             (Documentation and changelog)
|
+-- css/
|   +-- style.css         (External stylesheet)
|
+-- js/
|   +-- config.js         (Form URL, e-mail recipient, prices)
|   +-- site.js           (Accordion, tabs, lightbox, newsletter)
|   +-- products-data.js  (The product catalogue)
|   +-- products.js       (Search, filter, sort, modal)
|   +-- forms.js          (Validation and AJAX submission)
|   +-- map.js            (Leaflet map)
|
+-- docs/screenshots/     (Screenshots for the README)
|
+-- images/               (9 optimised images)
    +-- des-lux-play-logo.webp
    +-- favicon.png
    +-- luxury-handmade-stuffed-toys-hero.webp
    +-- des-j-sekgobela-founder-stitching.webp
    +-- standard-collection-chestnut-teddy-bear.webp
    +-- custom-purple-unicorn-plush.webp
    +-- custom-emerald-dragon-plush.webp
    +-- limited-edition-golden-lion-plush.webp
    +-- velvet-bunny-pastel-pink-plush.webp
    +-- cuddle-elephant-grey-plush.webp
    +-- silver-fox-limited-edition-plush.webp
    +-- des-lux-play-charity-donation-children.webp
```

Rationale for Structure

- **Root-level HTML files:** All HTML pages are placed in the root directory to simplify relative linking and so GitHub Pages can serve index.html. Navigation links (e.g., href="about.html") work without complex path traversal.
- **css/ folder:** Contains the external stylesheet (Part 2). Separating presentation from structure makes maintenance easier.
- **js/ folder:** Contains the Part 3 behaviour scripts, one per job. The single script.js planned in Part 1 was split so that each file is short and easy to find, and so that pages load only the code they need (for example, the map script loads only on the contact page).
- **images/ folder:** Consolidates all twelve visual assets, now named descriptively and saved as WebP for speed.
- **Search engine files:** robots.txt and sitemap.xml must be at the root so crawlers can find them.

3. Page Structure Descriptions

Each page follows a consistent semantic HTML5 structure comprising a skip link, header, navigation, main content area, sidebar, and footer. Every page has exactly one h1 heading, with h2 for sections and h3 for small headings.

Page	File	Content Structure
Homepage	index.html	Skip link > Header > Nav (with search) > Hero section (h1, hero image) > Standard Collection preview > Custom Creations preview > Limited Editions preview > Charity highlight > Sidebar (giveaway, animated charity tracker, newsletter form, quick links) > Footer
About Us	about.html	Header > Nav > Founder story (founder image + bio, h1) > Mission section > Vision section (text only) > Charity commitment (text only) > Gallery of 10 photos with lightbox dialog > Sidebar (maker profile, founded date, slogan) > Footer
Products	products.html	Header > Nav > Page banner > Catalogue (category buttons, sort menu, result count, grid of 7 toy cards built by JavaScript: 3 standard, 2 custom, 2 limited) > Good to Know tabs (sizes, ordering and delivery, care) > Sidebar (category links, price slider, recently viewed, custom order CTA) > Footer. A quick-view modal opens from each card.
Enquiry	enquiry.html	Header > Nav > Page banner > Custom order form (personal details fieldset > toy specifications fieldset > personalization fieldset > submit button) > Result panel (estimate, availability, e-mail button, shown after a valid submit) > FAQ accordion > Sidebar (order info, popular choices, help contact) > Footer
Contact	contact.html	Header > Nav > Page banner > Contact form (name, e-mail, optional phone, message type, message) > Result panel (compiled e-mail, shown after a valid submit) > Find Us section (nearest-store button, interactive map, Burgersfort HQ card, Polokwane store card) > Sidebar (quick contact, business hours, support info) > Footer
Not found	404.html	Header > Nav > Message with a link home > Sidebar quick links > Footer

Semantic HTML5 Elements Used

- **<header>** Contains the brand logo (des-lux-play-logo.webp) and company name, consistent across all pages.
- **<nav>** Two labelled navigation landmarks per page: the main menu with search, and a footer menu.

- **<main>** Wraps the primary content unique to each page; the skip link jumps to it.
- **<aside>** Contains sidebar widgets supplementary to the main content.
- **<footer>** Displays the footer menu, copyright information and location, consistent across all pages.
- **<section>** Groups related content blocks within the main area.
- **<article>** Used for self-contained content pieces such as product cards and location details.
- **<fieldset> and <legend>** Organise form elements into logical groupings on the enquiry page.
- **<address>** Marks up the store addresses on the contact page. (Added in Part 3.)
- **<dialog>** The quick-view modal and the photo lightbox, with built-in focus handling and Esc to close. (Added in Part 3.)
- **<figure> and <figcaption>** The enlarged photo and its caption inside the lightbox. (Added in Part 3.)
- **Tab and accordion markup** role=tablist, tab and tabpanel on the products tabs; buttons with aria-expanded on the FAQ accordion. (Added in Part 3.)

4. Navigation System

The navigation system is designed to be intuitive, consistent, and accessible. All pages share an identical navigation structure, ensuring users can orient themselves regardless of their current location within the site.

Navigation Menu Structure

```
+-------------------------------------------------------------+
|  [LOGO] Des-Lux Play                                        |
+-------------------------------------------------------------+
|  HOME | PRODUCTS | ABOUT | ENQUIRY | CONTACT   [Search] [Go] |
+-------------------------------------------------------------+
|  Footer: Home  Products  About  Enquiry  Contact  Instagram |
+-------------------------------------------------------------+
```

Navigation Features

- **Consistency:** The same five links appear in the same order on every page, preventing user disorientation.
- **Functional Links:** All navigation links were tested to ensure they load the target page. A link checker run in Part 3 found no broken internal links.
- **Self-Referencing:** Each page links to itself, which is standard practice and ensures link uniformity. The link for the current page is now marked with aria-current="page", so it is highlighted and announced by screen readers.
- **Search Integration (implemented in Part 3):** The search box at the right end of the navigation bar is now fully working. On the products page it filters the catalogue as the visitor types. On every other page it opens products.html?q=... with the results already filtered. It works even without JavaScript, and empty searches are blocked.
- **Footer navigation (added in Part 3):** A footer menu repeats the five links plus Instagram on every page, so visitors never have to scroll back up.

- **Skip link (added in Part 3):** A skip link lets keyboard users jump past the menu straight to the content.
- **Accessibility:** Navigation links use clear, descriptive text (HOME, PRODUCTS, ABOUT, ENQUIRY, CONTACT) rather than vague labels. The two menus are labelled Main and Footer so screen reader users can tell them apart.
- **Filtered links:** Sidebar category links open the catalogue already filtered (for example products.html?category=custom), and the address can be shared.

Two-Column Layout

Each page employs a two-column layout where the main content occupies approximately 70% of the width and the sidebar occupies 30%. This layout mirrors the provided website diagram and ensures that supplementary information (widgets, promotions, quick links) remains visible without competing with primary content for attention. Below 900 px wide the sidebar moves under the main content so that the pages stay readable on tablets and phones, and the pages were checked for sideways scrolling at 390 px.

5. References

- W3Schools. 2024. HTML5 Semantic Elements. Available at: <https://www.w3schools.com/html/html5_semantic_elements.asp> (Accessed: 13 August 2026).
- MDN Web Docs. 2024. HTML: HyperText Markup Language. Available at: <https://developer.mozilla.org/en-US/docs/Web/HTML> (Accessed: 13 August 2026).
- Google Maps Platform. 2024. Embed a Map. Available at: <https://developers.google.com/maps/documentation/embed> (Accessed: 13 August 2026). (Replaced in Part 3 by Leaflet, listed below; the map now also links to Google Maps.)
- Formspree. 2024. Form backend for HTML forms. Available at: <https://formspree.io> (Accessed: 13 August 2026).

Added in Part 3 (add your own access date to each):

- Agafonkin, V. n.d. Leaflet: an open-source JavaScript library for interactive maps. Available at: <https://leafletjs.com> (Accessed: 24 October 2026).
- W3C Web Accessibility Initiative. n.d. ARIA Authoring Practices Guide: Tabs and Accordion patterns. Available at: <https://www.w3.org/WAI/ARIA/apg/> (Accessed: 25 October 2026).
- MDN Web Docs. n.d. The Dialog element. Available at: <https://developer.mozilla.org/en-US/docs/Web/HTML/Element/dialog> (Accessed: 25 October 2026).
- Sitemaps.org. n.d. Sitemaps XML format. Available at: <https://www.sitemaps.org/protocol.html> (Accessed: 26 October 2026).
- GitHub. n.d. GitHub Pages documentation. Available at: <https://docs.github.com/pages> (Accessed: 26 October 2026).